

Fig 2 and 3 so they are more stand alone (would require redefining h and R_0 , etc)

Rebuilding-Time Based Reference Points: A Forward-Looking Risk-Equivalent Framework for Fisheries Management

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Author contributions: All authors designed the study, LTK and TKM performed the statistical analyses and led the writing of the manuscript. All authors contributed to the writing and review of the article and approved the submitted version.

Abstract

Biomass-based reference points are used to define stock status and set exploitation levels, but substantial variation across data-tiers and assessment frameworks can create inconsistent precaution. Risks should be equivalent across stocks, e.g. by maintaining comparable probabilities of failing to rebuild within agreed horizons. This study introduces B_{rebuild} , the lowest biomass from which a stock can rebuild to a target within an agreed timeframe under a specified fishing scenario, and T_{rebuild} , the time required to rebuild from a given state. B_{rebuild} can serve as a limit or trigger in harvest control rules; both metrics depend on the chosen target, horizon, and fishing mortality during rebuilding, F_{rebuild} . Using age-structured projections for 82 Northeast Atlantic stocks, we report conditional T_{rebuild} from assessment numbers-at-age under $F_{\text{rebuild}} = 0$, and show that recovery is faster at higher SPR_0 and steepness, with little contrast between Beverton–Holt and Ricker forms once these are accounted for. The approach shifts emphasis from recruitment impairment to rebuilding capacity, makes the trade-off with foregone yield explicit, and provides a common currency for risk-equivalent advice across assessment models and data tiers.

Keywords: reference points; harvest control rules; stock recovery; stock–recruitment relationships; production functions; risk-equivalent management; fisheries stock assessment

1 Introduction

Fisheries management aims to optimise yield, safeguard stock productivity, and maintain ecosystem health while balancing the trade-offs between biological, economic, and social objectives (Garcia, 1996). Achieving these potentially conflicting goals requires agreement on actionable objectives, including desired states with associated time frames, and the definition of target, limit, and threshold or trigger reference points (Rosenberg and Restrepo, 1996). Targets describe desirable long-term conditions for the stock, while limits define bounds intended to prevent unacceptable risks (Caddy and Mahon, 2004). In many frameworks, limit reference points represent biomass levels (e.g. B_{lim}) below which serious biological consequences are expected, whereas threshold or trigger points (e.g. B_{trigger}) are used to initiate pre-emptive action to avoid breaching those limits. When biomass is above a trigger reference point, exploitation can proceed at the target harvest rate; as biomass declines below it harvest rates become progressively more restrictive, while if the stock falls below B_{lim} a fishery closure may be required until the stock is rebuilt.

Figure 1 illustrates how these reference points are used as operational control points in a harvest control rule (HCR).

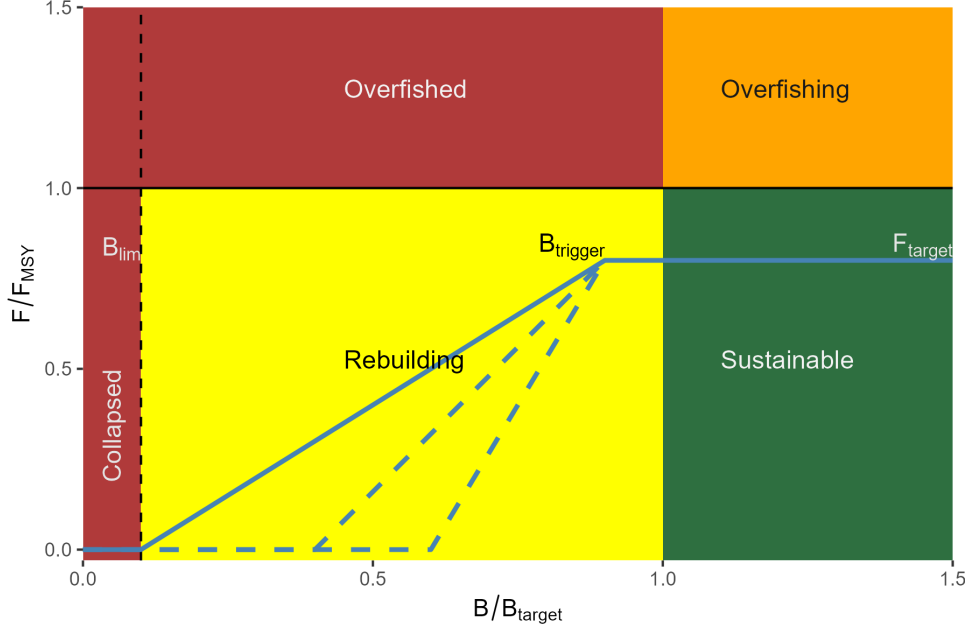


Figure 1: Generic harvest control rule showing the relationship between relative fishing mortality (F/F_{target}) and relative biomass (B/B_{target}). Key operational reference points include B_{lim} , $B_{trigger}$, and B_{target} . Management regions are indicated as target, rebuilding and high-risk zones.

Achieving sustainability requires keeping stocks above levels at which productivity is impaired. In practice, how impairment is determined depends on the assessment: data-rich age-structured analyses typically rely on stock–recruitment relationships, whereas biomass-based assessments use surplus production functions. Differences in data quality and model structure can therefore produce inconsistent precaution across frameworks. There is therefore a need for reference points that can be calibrated to deliver risk-equivalent advice, maintaining similar probabilities of impaired productivity (Fischer et al., 2023; Walker et al., 2023; Roux et al., 2022). Here, we define “risk” as the probability of undesirable outcomes, such as falling below a limit biomass reference point or failing to rebuild to a target within a specified time horizon. “Risk equivalence” is then defined as maintaining comparable levels of such risk across stocks and assessment tiers, accounting for differences in uncertainty. This typically requires more precautionary catch advice for assessments with greater uncertainty.

This study introduces a framework based on two complementary rebuilding-based quantities: $B_{rebuild}$ (rebuilding biomass) and $T_{rebuild}$ (rebuilding time). Depending on how strictly it is applied, $B_{rebuild}$ can play the role of B_{lim} (a level to avoid) or $B_{trigger}$ (an earlier warning sign), as in Figure 1. $T_{rebuild}$ describes the time required to reach the cho-

sen target from a given state and can be used to evaluate the consequences of alternative reference-point choices by asking, for example, how long rebuilding, and a reduction in yield, would be if biomass were allowed to fall to a given biomass level under a given harvest control rule. It can also be reported dynamically as an annually updated indicator alongside spawning biomass. B_{rebuild} and T_{rebuild} are designed to complement, not replace, current objectives and can be evaluated alongside yield, variability, and socio-economic metrics. The framework is fully compatible with F-based rebuilding plans, where fishing mortality rates are the primary management control, and T_{rebuild} serves as a performance metric characterising outcomes under stated policies rather than a prescriptive constraint.

Operationally, B_{rebuild} and T_{rebuild} can enhance and extend current practice. For example, in International Council for the Exploration of the Sea (ICES) advice, B_{rebuild} can serve as a stock-specific alternative to generic fractions of B_0 when setting B_{lim} or B_{trigger} under the MSY approach. In U.S. Magnuson–Stevens practice, rebuilding time is already included in some cases in the definition of overfished thresholds, and T_{rebuild} can be reported as a performance metric within rebuilding plans and harvest strategies (National Research Council, 2014). More generally, both quantities can play dual roles: as operational control points in HCRs, and as performance metrics in Management Strategy Evaluation (MSE) for comparing management procedures across stocks and data tiers (Punt et al., 2016; Cox et al., 2013; Winker et al., 2025).

Rebuilding time frames are policy choices, and the U.S. National Research Council (National Research Council, 2014) concluded that rebuilding plans should allow managers to select target fishing mortality based on biological productivity and management objectives. U.S. guidelines, however, can create problematic discontinuities: for example, if the minimum rebuilding time (T_{min} , calculated with zero fishing) crosses 10 years, the maximum allowable time (T_{max}) jumps from 10 years to T_{min} plus one generation time, which could double or triple the allowable rebuilding period and paradoxically permit higher fishing mortality for more depleted stocks. This highlights the need for more flexible approaches.

B_{rebuild} and T_{rebuild} can provide a common currency across assessment models and frameworks, shifting the focus from short-term forecasts toward medium-term recovery trajectories. They can also explicitly account for demography if based on age-structured

assessments and associated indicators (e.g. Griffiths et al., 2024). In this paper we define T_{rebuild} conditional on the current state vector, i.e. projected from the current assessment numbers-at-age. This provides a forward-looking indicator of rebuilding potential that reflects stock-specific year-class strength.

2 Materials and Methods

We first describe the derivation of B_{rebuild} and T_{rebuild} and then evaluate their application using a database of Northeast Atlantic stocks. Although the terminology used throughout is specific to the Northeast Atlantic and the ICES, analogous terminology is used in other jurisdictions.

2.1 Stock Assessment Database

For the case study we used 82 Northeast Atlantic stocks from the ICES FLR stock assessment database (e.g. Cardinale et al., 2024; Griffiths et al., 2024). All stocks are assessed using age-based methods (12 different frameworks, primarily SAM (Nielsen and Berg, 2014) and Stock Synthesis (Methot Jr and Wetzel, 2013)) with final data years from 2019 to 2022. Stocks are referred to by their ICES stock IDs (<https://sid.ices.dk>), for more details on the database see Cardinale et al. (2024); Griffiths et al. (2024); Winker et al. (2026)

2.2 B_{rebuild} and T_{rebuild}

B_{rebuild} is a biomass reference point: for example a level of spawning biomass used to trigger management action or a level to be avoided with high probability (e.g. as a limit or a trigger in an HCR; Figure 1). It could be set based on biological risk as currently done for B_{lim} in many frameworks, or based on policy considerations such as the lowest biomass still compatible with reaching an agreed target within an agreed recovery horizon under an agreed fishing mortality during rebuilding. T_{rebuild} is the corresponding time: the years (or generation times) needed to reach that target from a specified starting state, such as the current assessment or a predefined biomass reference point.

These definitions must be agreed and fixed before any evaluation, and require three choices to be made:

- B_{target} : The target biomass for recovery, typically expected biomass at Maximum Sustainable Yield (B_{MSY}), or another agreed biomass objective for example, an ecosystem-based target, where B_{MSY} is not appropriate.
- **Recovery horizon**: The planning period within which rebuilding to B_{target} is required (for example, n years or a multiple of generation time). This choice allows calibration of B_{rebuild} : a shorter horizon implies a higher B_{rebuild} .
- F_{rebuild} : Fishing mortality during recovery, set by an HCR or held constant. Here we use $F_{\text{rebuild}} = 0$ only as a standardised biological benchmark for comparing stocks. In practice a HCR would be used, and so F would increase as biomass rebuilds; for example, from 0 to F_{target} between B_{rebuild} and B_{trigger} .

In this study metrics are computed from single stock age-structured assessment estimates, but could be based on biomass based assessments for data-limited or moderate situations, or ecosystem models.

Generation time (τ) to express horizons or recovery times in biological units, is the mean age of spawners weighted by maturity and weight-at-age:

$$\tau = \frac{\sum_a a \cdot \exp\left(-\sum_{j=1}^{a-1} M_j\right) \cdot m_a \cdot w_a}{\sum_a \exp\left(-\sum_{j=1}^{a-1} M_j\right) \cdot m_a \cdot w_a} \quad (1)$$

where a is age, M_j is natural mortality at age j , m_a is maturity-at-age, and w_a is weight-at-age (Heppell et al., 2000; Waples, 2016).

Although we focus on spawning stock biomass as the state variable, the same definitions apply to any management target (e.g. age structure or spatial distribution).

B_{rebuild} offers several advantages: i) **empirical validation**, it is based on recovery trajectories that can be monitored and responses adapted; ii) **risk equivalence**, calibration can be conducted to ensure consistent probabilities of stock recovery across data tiers within hierarchical assessment frameworks; iii) **adaptability**, as life-history traits, spatial

processes, and environmental drivers can be explicitly incorporated; and iv) **trade-offs**, managers and stakeholders can consider economic and social objectives when setting the recovery horizon and F_{rebuild} in line with agreed risk tolerance.

2.3 Derivation

Rebuilding metrics were obtained from age-structured forward projections, conditional on the current state vector. Recent age-specific mass, maturity, selectivity, and natural mortality, as used to define B_{MSY} and related benchmarks (Sissenwine and Shepherd, 1987; Mangel et al., 2013; Kell et al., 2007), were assumed over the projection horizon. Equilibrium yield–biomass curves (production functions) were also derived and are shown in Supplementary Figure S5.

The recovery trajectories used for T_{rebuild} follow the age-structured population equations in Supplementary Methods Section S1: numbers-at-age are updated annually by survival and recruitment, recruitment follows the fitted stock–recruitment relationship (Supplementary Figure S4).

Conditional projections. The main analysis is conditioned on the final assessment year: initial numbers-at-age are taken from the assessment, $F_{\text{rebuild}} = 0$ is applied as a standardised biological benchmark (an HCR-based F could be substituted), and recruitment variability is incorporated by Monte Carlo resampling of stock–recruitment residuals. T_{rebuild} is the time to B_{MSY} (or another agreed B_{target}) from that state. For reading B_{rebuild} against an agreed horizon, age-structured trajectories from a grid of relative biomass levels (equilibrium initial numbers-at-age at each depletion) are reported in Supplementary Figure S3; those exploratory curves use the same dynamical equations and are not a biomass-only production-function projection.

B_{rebuild} depends on assumptions about recruitment dynamics; we therefore conducted a sensitivity analysis for alternative stock–recruitment relationships. Three parameterisation approaches were applied to each stock: (1) estimating both steepness (h) and unfished recruitment (R_0) within bounded ranges; (2) estimating R_0 with life-history-informed steepness priors (FishLife; Thorson et al., 2023); and (3) estimating steepness with R_0 fixed at the breakpoint of a segmented-regression (hockey-stick) model representing the recruitment-impairment threshold as estimated by ICES. Projections were

open-loop (no feedback between realised status and the control rule).

To summarise drivers of rebuilding time across the 82 stocks and alternative stock–recruitment scenarios, we fitted a generalised linear mixed model (GLMM) for T_{rebuild} to B_{MSY} , with stock as a random effect and steepness, $\log(\text{SPR}_0)$, mean spawner weight, production-curve shape, SRR form, and estimation scenario as fixed effects (Kristensen et al., 2016). The purpose is empirical multi-stock synthesis for calibration and risk-equivalence, not a new theoretical result: full model specification, covariate definitions (Table S1), coefficients (Table S2), and the predictor-correlation figure (Figure S1) are in Supplementary Materials (Section S2); diagnostic, effect, and elasticity plots are in the companion GitHub repository.

3 Results

Stock–recruitment fits for the stocks vary substantially where contrast in SSB–recruitment is limited (Supplementary Materials). In ICES practice a segmented regression (hockey-stick) SRR is used mainly to identify a breakpoint for an operational control point rather than as a biological hypothesis (Mesnil and Rochet, 2010). Continuous Beverton–Holt and Ricker forms were therefore evaluated in the T_{rebuild} figures, with hockey-stick comparisons in Supplementary Figures S4–S5. Conditional T_{rebuild} for $F_{\text{rebuild}} = 0$ is summarised in Figure 2 for stocks with final-year SSB $< 0.80 B_{\text{MSY}}$. Monte Carlo medians that resample recruitment residuals while preserving the observed age structure are compared with deterministic predictions at the same depletion. Agreement is generally good, close to the $y = x$ line, with uncertainty bands reflecting recruitment variability; larger deviations occur for some stock–scenario combinations, particularly under Ricker dynamics where a strong year class can substantially change recovery time.

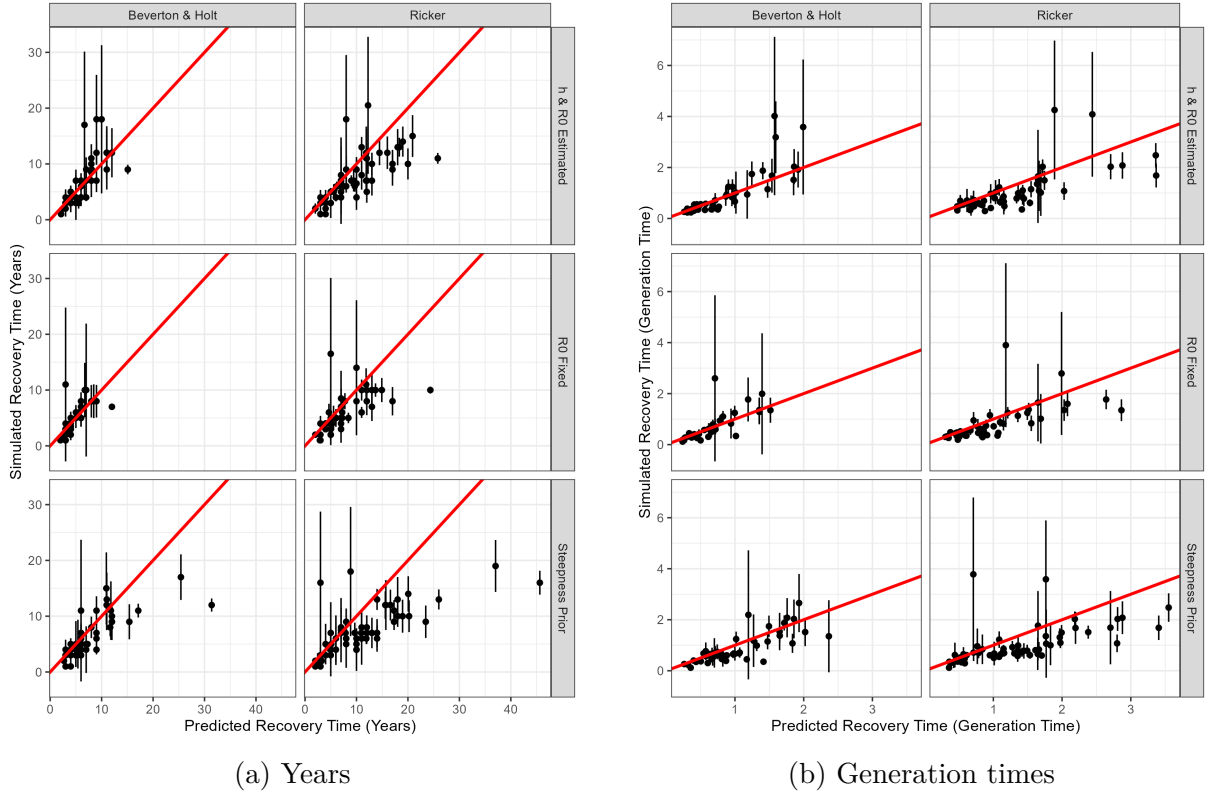


Figure 2: Conditional T_{rebuild} under $F_{\text{rebuild}} = 0$: deterministic age-structured prediction at the stock's final-year depletion (horizontal) versus Monte Carlo median T_{rebuild} from projections conditioned on assessment numbers-at-age (vertical; error bars ± 2 SD across 100 replicates). Each point is one stock–scenario combination. The red line is $y = x$. Panels show calendar years (a) and generation times τ (b).

Recovery times by stock (lines) plotted against relative biomass are shown in Figure 3, with illustrative management horizons at five years and one generation time. Stocks above the horizon reference line cannot rebuild within 5 years or one generation time under the assumed fishing scenario, providing a direct, monitorable indicator of whether current status is compatible with agreed recovery objectives. Variability across stocks is lower when T_{rebuild} is expressed in generation times.

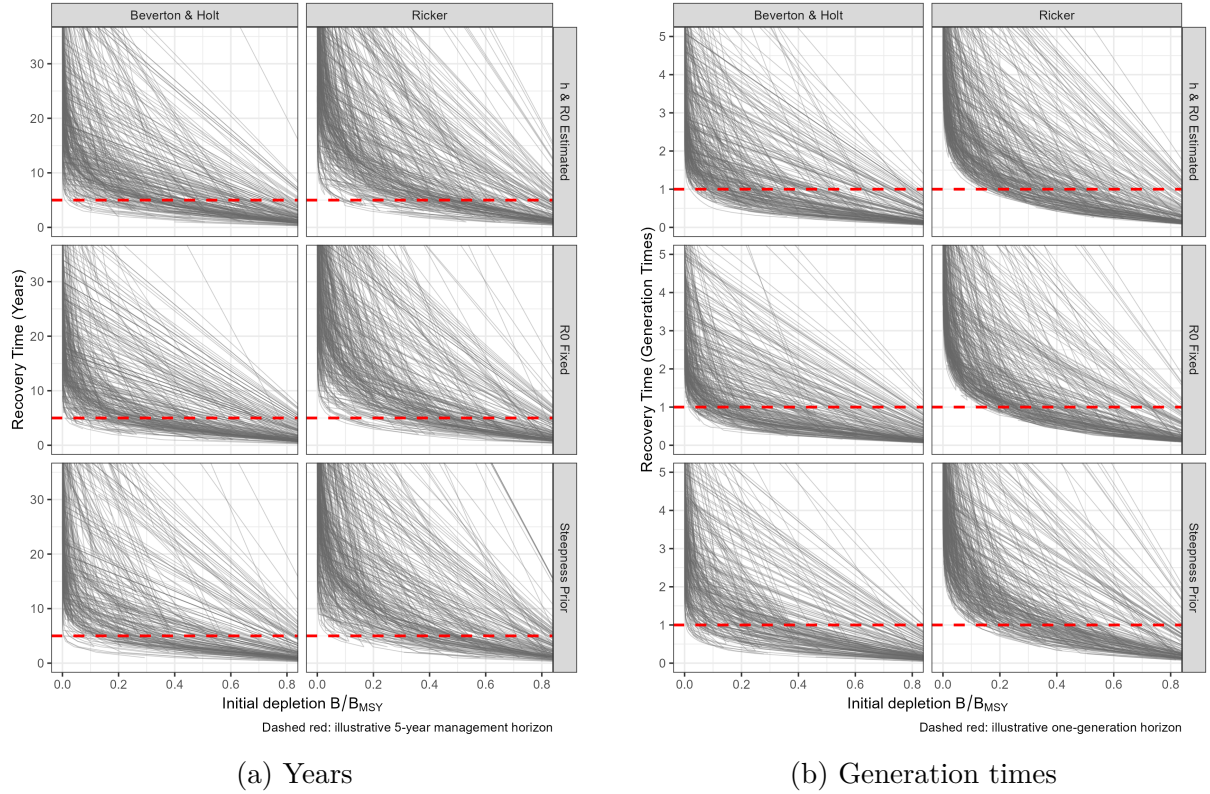


Figure 3: Conditional T_{rebuild} versus final-year stock status ($\text{SSB}/B_{\text{MSY}}$) for stocks below $0.80 B_{\text{MSY}}$, under $F_{\text{rebuild}} = 0$. Bars summarise Monte Carlo projections that resample recruitment residuals from the fitted stock–recruitment relationship. Dashed horizontal lines mark illustrative horizons of five years (a) and one generation time (b).

3.1 Factors influencing rebuilding time

Across stocks and scenarios, shorter rebuilding times were associated with higher SPR_0 and higher steepness (Table S2; Supplementary Section S2). The GLMM analysis summarised T_{rebuild} to B_{MSY} from conventional B_{lim} and B_{trigger} starting points, not the conditional current-status runs in Figures 2–3. Ricker versus Beverton–Holt form had little additional effect once these covariates were included. Production-curve shape (B_{MSY}/B_0) is not treated as an independent driver of recovery: plotting shape against steepness (Figure S2) shows a moderate negative association under Beverton–Holt ($r \approx -0.53$)—so apparent shape effects largely track steepness for that form—but only a weak association under Ricker ($r \approx -0.12$; overall $r \approx -0.21$). We therefore emphasise SPR_0 and steepness as the biological drivers.

4 Discussion

B_{rebuild} and T_{rebuild} provide a framework that shifts emphasis from recruitment impairment to forward-looking rebuilding capacity (Fischer et al., 2023; Walker et al., 2023; Roux et al., 2022). By explicitly linking biomass thresholds to recovery trajectories and management time frames, the approach focuses on what should be monitored in well-managed fisheries rather than on impairment events that should only be rarely observed. Risk equivalence across stocks and assessment tiers can be achieved through calibration so that advice reflects comparable probabilities of failing to rebuild within agreed horizons. Operationally, B_{rebuild} and T_{rebuild} can be used in reporting and MSE as transparent cross-stock metrics (Punt et al., 2016; ?). The framework is fully compatible with F -based harvest control rules where fishing mortality increases as biomass rebuilds. The variance of T_{rebuild} reflects both process error and life-history effects on recovery capacity; when variance is large, it indicates limited forecasting skill and should lead to more restrictive management.

Results showed that equilibrium-based B_{rebuild} provides a robust description of rebuilding dynamics across stock–recruitment assumptions. The GLMM results indicated that recovery is faster when SPR_0 and steepness are high and slower when production curves are more left-skewed. The choice between Ricker and Beverton–Holt models had little effect once these biological covariates were included, indicating that rebuilding capacity depends primarily on reproductive potential and the left-hand side of the production function. Rebuilding can vary substantially between stocks depending on life history. Long-lived, slow-growing species may require higher biomass limits to ensure recovery within the same management cycle as more resilient stocks. This highlights why generic biomass-based limits (e.g., $0.2B_0$) are not risk-equivalent across stocks and supports tailoring management thresholds to stock-specific recovery capacity. For a given set of life-history parameters, B_{rebuild} provides a more robust basis for limit reference points across plausible stock–recruitment choices than simple B_0 -based fractions.

Equilibrium-based B_{rebuild} is tied to an average age structure and therefore summarises rebuilding capacity for a representative depleted state; conditional T_{rebuild} uses the full numbers-at-age vector and tracks realised recovery timelines from the assessment year. Expressing T_{rebuild} in generation times (Heppell et al., 2000; Waples, 2016) provides a

biologically interpretable scale alongside calendar years, with reduced variability due to incorporation of stock-specific characteristics. Stochastic projections showed that actual recovery times vary around equilibrium predictions, particularly when year-class strength fluctuates, but the qualitative importance of life-history drivers remains consistent. Preserving favourable age structure may accelerate rebuilding relative to equilibrium averages, suggesting that managing for age diversity improves recovery performance.

The framework addresses two key limitations of current rebuilding approaches. Firstly, achieving biomass targets inherently requires rebuilding capacity, yet many jurisdictions artificially separate “normal” and “rebuilding” management regimes with different HCRs for depleted versus non-depleted stocks (National Research Council, 2014). This creates abrupt changes in fishing mortality when stocks cross classification thresholds. Because B_{rebuild} and T_{rebuild} explicitly link targets to the time and trajectory needed to reach them, the framework unifies target-setting and rebuilding within a single, continuous approach. Secondly, the framework separates biological assessment from policy decisions. Scientists provide B_{rebuild} and the distribution of T_{rebuild} conditional on specified policy choices (target biomass, rebuilding horizon, fishing scenario), while managers select among alternatives based on socio-economic considerations and risk tolerance (National Research Council, 2014).

T_{rebuild} is also valuable as a dynamic performance indicator serving as a common currency across assessment tiers—age-structured, biomass-based, and MSE—so advice is compared on recovery capacity rather than model-specific biomass ratios. Biomass fractions derived from SRR breakpoints or virgin biomass (e.g. $0.2B_0$) are not risk-equivalent across stocks, whereas B_{rebuild} can be tailored to recovery capacity and then monitored to help with validation (were outcomes as predicted by the model). Reporting T_{rebuild} as standard output alongside SSB would provide transparent communication of recovery progress and shift focus toward medium-term recovery trajectories. Presenting the full distribution rather than a single point estimate makes rebuilding uncertainty explicit and supports informed policy trade-offs. Under current U.S. guidelines, rebuilding plans are designed to achieve target biomass with at least 50% probability, meaning approximately half are expected to miss their target time even when implemented as designed (National Research Council, 2014). By showing how rebuilding probability varies with time horizon and fishing mortality, managers can evaluate whether to accept this standard or adopt more conservative

thresholds. Rebuilding choices also have explicit social and economic consequences. The horizon and F_{rebuild} determine how long catch must be reduced: foregone yield is the tangible cost of precaution. Framing limits in recovery time makes that trade-off transparent when managers set B_{rebuild} , horizons, and risk tolerance.

B_{rebuild} can be incorporated into MSE frameworks by defining it as a risk-equivalent reference point within the operating model. For data-limited stocks, B_{rebuild} can be calculated using surplus production models, with benchmarks risk-calibrated against age-structured assessments to ensure consistent advice across data tiers. Rebuilding based metrics also extend beyond single-stock biomass targets: replacing B_{target} with an EBFM objective (e.g. age structure, occupancy, or a multispecies constraint), or computing trajectories from ecosystem models, yields rebuilding metrics for those objectives. Spatial contraction and expansion can affect rebuilding dynamics and should be considered when applying B_{rebuild} and T_{rebuild} . More generally, these metrics can be evaluated for targets defined on age structure or spatial distribution by replacing B_{target} with the corresponding objective. Many existing priors are derived on a per-recruit basis, which may not fully capture that depleted populations must recolonise suitable habitats (Cardinale et al., 2025). Future work should address whether abundance–occupancy relationships are path-dependent (Kidd et al., 2025), which has implications for interpreting spatial contractions and designing HCRs that react differently when stocks are rebuilding versus declining. Sensitivity analysis could be expanded to include broader sources of uncertainty (e.g., Allee effects; Albertsen et al. 2025). The current analysis assumes zero fishing mortality during rebuilding; future work should explore how different fishing mortality scenarios affect trajectories.

Operationally, scientific advice can report B_{rebuild} and the distribution of T_{rebuild} under stated targets, horizons, and fishing scenarios, while managers choose among alternatives. The present analysis used $F_{\text{rebuild}} = 0$ as a biological benchmark and as an illustrative management example; alternative fishing scenarios, spatial recolonisation, and related uncertainties remain for further work (Cardinale et al., 2025; Albertsen et al., 2025).

Assuming a hockey-stick stock–recruitment curve, as generally done by ICES (?), results in truncated production functions that may underestimate rebuilding capacity. Since the hockey-stick constrains expected recruitment to the mean after the breakpoint, it fails to

capture the range of biological possibilities. For many ICES stocks, the estimated hockey-stick breakpoint lies at or very close to B_{MSY} , which would imply a limit reference point essentially equal to the target. In contrast, B_{rebuild} and T_{rebuild} are linked to rebuilding from depleted states and therefore define values strictly below B_{MSY} for a given time horizon.

5 Conclusion

This study introduced B_{rebuild} and T_{rebuild} as forward-looking reference points that shift emphasis from recruitment impairment to rebuilding capacity. Results showed that rebuilding dynamics are driven primarily by reproductive potential (SPR_0) and steepness, and once these were taken into account that the functional form of the SRR was less important. This highlights the importance of low-biomass dynamics (Hutchings, 2015; Perälä et al., 2022).

Conditional T_{rebuild} , projected from the assessment numbers-at-age, provides a dynamic performance indicator of recovery and a common currency comparable across assessment models. Because the recovery horizon and F_{rebuild} also determine how long catch must be reduced, the framework makes the trade-off between biological risk and foregone yield explicit for managers and stakeholders.

Like other reference points from fitted life-history and stock–recruitment models, these quantities rely on stationarity assumptions. Their management value is that conditional T_{rebuild} can be updated with the state vector to monitor progress toward targets and to support adaptive adjustment of advice. Used as a shared performance metric, it can help harmonise risk advice across data-rich and data-limited stocks (Silvar-Viladomiu et al., 2022; Hordyk et al., 2015; Punt et al., 2016; Winker et al., 2025; Roux et al., 2022).

6 Acknowledgements

L.T.K., M.C. and C.G. were partially funded via a FORMAS - a Swedish Research Council for Sustainable Development grant (grant number: 2023- 01392 awarded to C.G.). We also thank Andre Punt and an anonymous reviewer for their insightful comments on a

previous version of the paper.

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